

ResearchGraph AI: A Semantic Academic Research Intelligence Platform Using Large Language Models and Knowledge Graphs

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Abstract—The rapid growth of scientific publications has made academic literature increasingly difficult to explore efficiently. Researchers often need to search many papers, compare methodologies, identify datasets and models, trace relationships, and determine potential research gaps. Conventional academic search platforms are effective for paper discovery but generally present documents as independent results, while traditional Retrieval-Augmented Generation (RAG) systems primarily retrieve semantically similar text chunks. ResearchGraph AI is proposed as a semantic academic research intelligence platform that combines Large Language Models (LLMs), vector retrieval, Knowledge Graphs, and Graph Retrieval-Augmented Generation (GraphRAG). The system collects research papers from scholarly sources such as OpenAlex and arXiv, extracts and preprocesses document content, creates semantic embeddings, and stores the embeddings in a vector database. At the same time, research entities such as papers, authors, datasets, algorithms, models, and research domains are extracted and represented as nodes and relationships in a Neo4j Knowledge Graph. When a user submits a natural-language query, the system performs semantic retrieval from the vector store and relationship-based retrieval from the Knowledge Graph. The retrieved evidence is combined and supplied to an LLM to generate a grounded, citation-supported response. The proposed platform is intended to support literature review, paper comparison, research relationship exploration, and research-gap identification, reducing the manual effort required for academic research.

Index Terms—academic research intelligence, semantic search, Knowledge Graph, GraphRAG, Retrieval-Augmented Generation, vector database, Large Language Models.

I. INTRODUCTION

The amount of scientific literature available to researchers has increased rapidly because of digital publishing, open repositories, and continuous publication of conference and journal articles. Although this growth improves access to knowledge, it also creates a practical problem: a researcher may need to examine a large collection of papers before understanding the main methods, datasets, models, findings, and unresolved problems in a topic. Searching for papers is only the first step; connecting the information distributed across papers is often the more difficult task.

Academic search platforms provide useful indexing, citation information, and paper discovery. However, a researcher who

wants to answer a relationship-oriented question may still have to open several papers and manually connect the evidence. For example, a query about which datasets were used by different papers for evaluating a particular model requires both semantic understanding of paper content and explicit relationships among papers, methods, and datasets.

ResearchGraph AI is proposed to address this problem through a combination of semantic retrieval and graph-based reasoning. The platform treats research literature as both textual evidence and structured research knowledge. Textual information is represented through embeddings for semantic search, while important research entities and their relationships are represented in a Knowledge Graph. A Large Language Model then uses evidence retrieved from both representations to produce a grounded response.

II. RELATED WORK

A. Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) combines language-model generation with retrieval from an external knowledge source. Ma et al. survey the development of RAG approaches and discuss retrieval, augmentation, and generation as important components for improving the usefulness of LLM responses in knowledge-intensive tasks [3]. Procko and Ochoa further examine GraphRAG as an extension of RAG in which Knowledge Graphs are incorporated to provide more structured information for retrieval and generation [2]. These studies establish the importance of retrieving external evidence instead of relying only on the information encoded in an LLM.

B. Large Language Models and Knowledge Graphs

Pan et al. present a roadmap for unifying LLMs and Knowledge Graphs and categorize existing approaches into KG-enhanced LLMs, LLM-augmented KGs, and synergized LLM-KG systems [1]. Their work shows that Knowledge Graphs can provide structured factual information and support reasoning, while LLMs can assist with graph construction, completion, question answering, and natural-language generation. This combination is particularly relevant to ResearchGraph AI

because the proposed system uses both an LLM and a Knowledge Graph as complementary components.

C. GraphRAG and Text-Chunk Knowledge Graphs

Procko and Ochoa survey Graph Retrieval-Augmented Generation and highlight the use of Knowledge Graphs to organize information and reduce noise in retrieved context [2]. Xiao et al. proposed TKG-RAG, a Retrieval-Augmented Generation framework that incorporates a text-chunk Knowledge Graph into the retrieval process [6]. Their approach is relevant to ResearchGraph AI because the proposed platform similarly connects document-level information with graph-based relationships rather than relying exclusively on independent vector chunks.

Wang et al. introduced XGraphRAG, an interactive visual-analysis framework for examining GraphRAG pipelines and tracing retrieval results through the graph-based retrieval process [5]. This work emphasizes the importance of understanding and analyzing GraphRAG retrieval, which is also relevant to the proposed platform’s planned Knowledge Graph visualization and research exploration capabilities.

D. Knowledge Graph Representation and Question Answering

Wang et al. provide a survey of Knowledge Graph Embedding approaches and their applications, describing how entities and relationships can be represented in continuous vector spaces for downstream tasks [4]. Such representations provide a foundation for connecting research entities including papers, authors, datasets, algorithms, and models.

Lan et al. survey complex Knowledge Base Question Answering and discuss questions involving multiple subjects, compound relations, and multi-step reasoning [7]. These capabilities are important for academic research queries because a researcher may need to follow relationships across several entities. ResearchGraph AI therefore combines graph-based retrieval with semantic vector retrieval and LLM-based generation to support both textual and relationship-oriented research questions.

III. RESEARCH GAP AND MOTIVATION

The reviewed literature indicates that RAG is effective for retrieving domain-specific textual evidence, while Knowledge Graph based question answering is useful for representing structured relationships. However, these capabilities are often studied independently or in narrowly defined application domains. Academic research requires both capabilities because important questions may contain two different information needs: the relevant passages must be retrieved from papers, and relationships between entities must also be identified.

For example, consider the question: “Which datasets were used by papers that improved a particular model, and what evaluation metrics did they report?” A vector-only system can retrieve passages containing the model and datasets, but it may not explicitly establish the chain Paper $\rightarrow$ Uses Dataset $\rightarrow$ Evaluated With Metric. A graph-only system may represent the relationships but may not contain the detailed textual

explanation required for the answer. The proposed system therefore combines the two retrieval paths.

The main motivation of ResearchGraph AI is to reduce the manual effort involved in literature exploration while maintaining a connection between generated answers and the source papers. The system is designed as an academic research assistant rather than a replacement for scholarly databases. Retrieved papers remain the evidence source, while the LLM is used to organize and explain the retrieved information.

IV. PROPOSED SYSTEM

A. System Overview

The proposed system consists of a research-paper ingestion layer, document processing layer, semantic retrieval layer, Knowledge Graph layer, query processing layer, and response generation layer. Research papers are collected through scholarly APIs such as OpenAlex and arXiv. The paper metadata and available documents are processed to obtain clean textual content. The content is divided into manageable chunks and transformed into semantic embeddings.

The embeddings are stored in ChromaDB along with metadata such as paper identifier, title, section, and source. In parallel, entity extraction is applied to identify research concepts. Neo4j stores these entities and their relationships. The two databases therefore serve different purposes: ChromaDB provides similarity-based evidence retrieval, while Neo4j provides relationship-aware retrieval and graph traversal.

B. Research Paper Collection

The collection module obtains metadata such as paper title, authors, abstract, publication year, DOI, venue, and document URL. OpenAlex and arXiv are planned as primary sources because they provide programmatic access to scholarly metadata and research records. The ingestion process can be executed periodically so that newly available records can be added without rebuilding the entire knowledge base.

C. PDF Processing and Chunking

When a document is available as a PDF, the document-processing module extracts its text and removes unnecessary formatting artifacts. The cleaned text is segmented into chunks. Chunking is necessary because a complete research paper is generally too large to send to a language model as a single retrieval unit. Each chunk retains metadata linking it to the original paper so that retrieved evidence can later be traced back to the source.

D. Embedding and Vector Storage

Each processed text chunk is converted into a dense embedding using a suitable sentence-embedding model. The vector and its metadata are stored in ChromaDB. When a user asks a question, the same embedding process is applied to the query. Similarity search then returns the most relevant chunks. This allows the system to retrieve conceptually related passages even when the exact words in the query do not appear in the paper.

TABLE I
CORE SYSTEM COMPONENTS

Stage	Primary Component	Function
Collection	OpenAlex / arXiv	Paper metadata and documents
Semantic	ChromaDB	Embedding-based retrieval
Graph	Neo4j	Entity and relationship traversal
Generation	LLM	Grounded response generation

V. SYSTEM ARCHITECTURE

The overall architecture follows a modular pipeline. The user interacts with a React-based interface. A FastAPI backend receives requests and coordinates the ingestion, retrieval, graph, and LLM components. During ingestion, papers pass through PDF extraction, preprocessing, and chunking. The processed content branches into two paths: embedding generation for ChromaDB and entity/relationship extraction for Neo4j. During question answering, the backend performs vector retrieval and graph retrieval, aggregates the evidence, and sends the final context to the LLM.

VI. GRAPH RAG QUERY PROCESSING

A. Query Understanding

The user query is first normalized and converted into an embedding. The backend also identifies important entities or concepts that may correspond to nodes in the Knowledge Graph. This step determines whether the question requires only textual evidence or also requires relationship-based retrieval.

B. Vector Retrieval

The query embedding is compared with document embeddings in ChromaDB. The top- k most relevant chunks are returned together with their paper metadata. These chunks provide detailed textual evidence for the final answer.

C. Graph Retrieval

The identified entities are matched against Neo4j nodes. Cypher queries retrieve directly connected and, when necessary, multi-hop relationships. For example, a query can traverse from a research method to papers using the method, then to datasets used by those papers and the metrics reported for evaluation.

D. Context Aggregation and Generation

The vector results and graph results are merged and ranked or filtered before being passed to the LLM. The prompt instructs the model to answer from the supplied evidence and avoid unsupported claims. The generated response can include paper titles, authors, and source references to make the answer easier to verify.

E. Knowledge Graph Construction

The Knowledge Graph represents the structured side of the research collection. Typical node types include Paper, Author, Dataset, Method, Model, Research Domain, and Evaluation Metric. Relationships can include WRITTEN_BY, USES, TESTED_ON, CITES, IMPROVES, RELATED_TO,

and EVALUATED_WITH. A paper node can therefore be connected to its authors, methods, datasets, and related papers. These relationships enable graph traversal and support multi-step research questions, consistent with the requirements of complex Knowledge Base Question Answering [7].

F. Hybrid GraphRAG Retrieval

For a user query, the backend first creates a semantic representation of the query and retrieves relevant chunks from ChromaDB. Important entities in the query are also identified and matched with nodes in Neo4j. Cypher queries then retrieve related entities and relationships. The results from both paths are combined into a context package. This hybrid retrieval is the core of the proposed GraphRAG pipeline, following the broader direction of graph-enhanced retrieval described in the GraphRAG literature [2], [6].

G. LLM Response Generation

The final context is provided to an LLM such as Gemini or Llama 3 with instructions to answer using the retrieved evidence. The response-generation layer is expected to support question answering, paper summaries, comparisons, literature-review assistance, and research-gap exploration. Citations or paper identifiers are retained in the context so that the generated answer can be connected to its supporting sources.

VII. PROPOSED ALGORITHM

Algorithm 1. GraphRAG Query Processing

- 1) Input: natural-language query Q .
- 2) Generate embedding $E(Q)$.
- 3) Retrieve top- k semantic chunks from ChromaDB.
- 4) Extract or identify research entities from Q .
- 5) Match entities with Neo4j nodes and retrieve connected relationships using Cypher.
- 6) Combine vector evidence V and graph evidence G into context C .
- 7) Send C and Q to the LLM.
- 8) Generate response R with supporting paper references.
- 9) Output: citation-supported response R .

VIII. TECHNOLOGY STACK AND IMPLEMENTATION

A. Frontend and Backend

The frontend is planned with React to provide an interactive research interface for search, question answering, paper exploration, and Knowledge Graph visualization. FastAPI is planned as the backend framework because the application requires lightweight APIs for document ingestion, retrieval, graph queries, and LLM communication. Python also provides access to document-processing, embedding, and AI libraries.

B. Vector and Graph Databases

ChromaDB is used as the vector store because the system requires semantic similarity search over document chunks. Neo4j is used as the graph database because the project requires explicit storage and traversal of relationships. The separation allows each database to perform the type of retrieval for which it is best suited.

C. AI and Processing Components

LangChain can be used to coordinate document loading, chunking, retrieval, prompt construction, and LLM interaction. PyMuPDF can be used for PDF text extraction. An embedding model such as BGE can generate document and query embeddings. Gemini or Llama 3 can serve as the response-generation model depending on deployment constraints and API availability.

IX. EXPECTED OUTCOMES

The expected outcome is a working academic research intelligence platform that allows users to search and question a collection of research papers using natural language. The system is expected to provide semantically relevant evidence, expose relationships between research entities, and generate grounded responses. It can support literature-review preparation by summarizing papers and organizing related studies, assist in paper comparison by connecting methods and evaluation results, and help researchers explore possible research gaps by revealing areas with limited connections or repeated limitations.

The system will also provide a foundation for visual research exploration. A user can inspect a paper and move through its connected authors, methods, datasets, and related papers in the Knowledge Graph. This graph-based exploration complements normal document search and can make relationships easier to understand.

X. EVALUATION PLAN

The proposed system can be evaluated using a small benchmark of research-oriented questions. Evaluation categories may include direct fact questions, semantic retrieval questions, relationship-based questions, and multi-hop questions. Retrieval quality can be measured by checking whether the relevant papers or chunks are returned. Answer quality can be evaluated using correctness, faithfulness to retrieved evidence, citation correctness, and response relevance. A vector-only RAG baseline can be compared with the hybrid vector-plus-graph approach to determine whether graph retrieval improves relationship-oriented questions.

Additional evaluation can measure response latency, number of retrieved sources, and user-perceived usefulness. These measures will help determine whether the additional graph-processing stage provides a practical improvement over conventional semantic retrieval.

XI. CONCLUSION

ResearchGraph AI proposes a semantic academic research intelligence platform that combines vector retrieval, Knowledge Graph reasoning, GraphRAG, and Large Language Models. The central idea is to represent research literature in two complementary forms: textual evidence stored as semantic embeddings and structured knowledge stored as graph entities and relationships. This design allows the system to answer both content-oriented and relationship-oriented research questions. By connecting retrieved evidence to source papers, the platform aims to support literature review, paper comparison, research

exploration, and research-gap identification while reducing the manual effort involved in navigating large collections of academic publications.

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